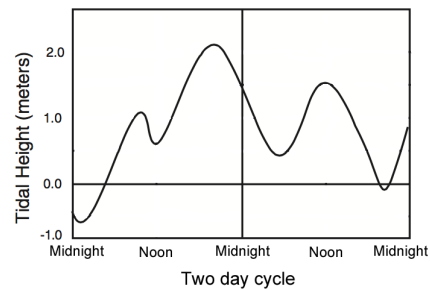
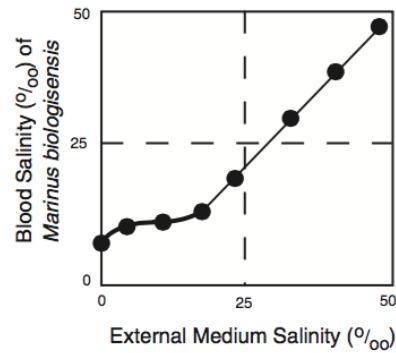


(3 pts) 1. Name the three zones of the rocky intertidal, from mean low tide to mean high tide.

(4 pts) 2. What is the name of the tidal pattern shown at right. Tell how you know.



(5 pts) 3. This figure shows the blood salinity of a marine worm relative to sea water. Is this worm mostly an osmoconformer or an osmoregulator? Tell how you know.



(5 pts) 4. Why is a stenohaline marine fish in a coastal plain estuary more likely to be located deeper in the water column the farther it swims up the estuary (towards the main river)?

(5 pts) 5. How do hypersaline patches form in New England marshes?

(5 pts) 6. Explain bioturbation and how it increases oxygen availability in the sediments of a soft-bottom intertidal community.

7. Write the name of your favorite band or musical artist.
(2 pts extra credit)

(8 pts) 8. Two days ago was a full moon. Tell whether that phase would have caused a neap tide or a spring tide, and then diagram and explain why.

(10 pts) 9. Explain how patch size affects whether a patch in the rocky intertidal will return to the original species or if a new species will be able to colonize and remain established. (HINT: California and Bay mussels.)

(15 pts) 10. Meiofauna refers to the small animals that live between sediment grains. Describe how the relative size (larger, smaller, etc.) of the meiofauna will change in a bar-built estuary from the river to the sand bar at the ocean. Tell why the size of the meiofauna changes.

(15 pts) 11. Explain how physical and competitive forces interact to explain zonation of the four major marsh plant species in a New England marsh.